ModelIt!

Teacher's Guide

Can Sound Travel Through Space?

Why Astronauts Need Radios and Explosions Are Silent in a Vacuum

Quick Reference

Grade Level	Grade 04L2
NGSS Standard	4-PS4-1, 4-PS4-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-PS4-1, 4-PS4-3

Develop a model of waves to describe patterns in terms of amplitude and wavelength. Generate and compare multiple solutions that use light or sound to transfer information.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a five-component model showing how vibration strength and medium density interact to determine wave speed, absorption, and loudness in different environments.
Disciplinary Core Idea	**PS4.A Wave Properties / PS4.C Information Transfer** — Waves can be made in water by disturbing the surface. Sound waves need a medium to travel. Information can be transferred through sound waves or electromagnetic waves.
Crosscutting Concept	**Cause and Effect** — Students investigate how changes in medium density cause predictable effects on wave speed and absorption, and discover the special case where removing the medium entirely eliminates sound.

CAST Testing Alignment

MULTIPLE CHOICE:

An astronaut in space watches a satellite explode 100 meters away. What does the astronaut hear?

CONSTRUCTED RESPONSE:

Using what you know about sound waves and media, explain why whales can communicate across hundreds of miles underwater but humans can only shout across a few hundred feet in air. Use the words 'medium,' 'wave speed,' and 'absorption' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Sound is something we experience every moment of our lives, yet most people never think about what sound actually IS. Understanding that sound requires a medium to travel changes how we think about the universe — most of space is completely silent, no matter how dramatic the events happening there.

What Sound Really Is

Sound is a mechanical wave — a vibration that passes through matter by bumping molecules into each other. When a drum is struck, the drum skin pushes air molecules forward, which bump into the next molecules, which bump into the next, creating a chain reaction that travels outward in all directions. This is fundamentally different from light, which is an electromagnetic wave that does NOT need molecules to travel.

Sound in Different Media

Sound travels at different speeds depending on the medium. In air at room temperature, sound moves at about 343 meters per second (767 mph). In water, it moves at about 1,480 m/s — over 4 times faster. In steel, it moves at about 5,960 m/s — over 17 times faster than in air. Why? In denser materials, molecules are packed closer together, so each molecule does not have to travel as far to bump into the next one. The chain reaction happens faster.

The Silence of Space

Space is an almost perfect vacuum — there are only about 1-10 atoms per cubic centimeter in interstellar space, compared to about 25 billion billion (2.5 x 10^19) molecules per cubic centimeter in air at sea level. With virtually no molecules, there is nothing to vibrate, and sound simply cannot exist. Astronauts communicate using radio waves, which are electromagnetic waves like light and can travel through vacuum. Every space movie that shows explosions with sound is scientifically wrong.

How Animals Use Sound in Different Media

Different animals have evolved to use sound in the medium that surrounds them. Whales use low-frequency sound waves that can travel thousands of miles through ocean water because water transmits sound efficiently with low absorption. Elephants communicate using infrasound (very low frequency sounds) that travel through the ground — they detect these vibrations through their feet. Bats use ultrasound (very high frequency) in air to navigate and find insects in the dark.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Vibration Strength and Medium Density as external components (what we control) and Wave Speed, Sound Loudness, and Wave Absorption as internal components that respond based on the medium and source.

E **Establish Relationships**

Students determine that Vibration Strength positively affects Loudness, Medium Density positively affects Wave Speed, Medium Density negatively affects Wave Absorption, and Wave Absorption negatively affects Loudness — creating competing pathways.

V **Visualize & Model**

Students build a five-component model showing how sound depends on both the source AND the medium, with two pathways affecting loudness (direct from source, and indirect through absorption).

E **Evaluate Outcomes**

Students run four scenarios (air, water, vacuum, strong vs. weak source) to observe how the medium dramatically changes sound behavior, with the vacuum scenario showing complete system shutdown.

R **Revise & Extend**

Students add distance or temperature to explore why sounds get quieter over distance and why sound behaves differently on hot vs. cold days.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Raise your hand if you have seen a space movie where something explodes with a big BOOM. What if I told you those movies are lying to you?"*

Watch For

Play a clip from a space movie with a loud explosion. Ask: Do you think that is what it would really sound like?

Slide 2: Learning Objectives

3 min

Say: *"Today we are going to figure out whether sound can travel through NOTHING — and why that matters for astronauts."*

Watch For

Have students read objectives aloud. Introduce the word 'medium' and 'vacuum' with real examples.

Slide 3: Big Question

3 min

Say: *"Here is the question: space has almost nothing in it. Can sound travel through nothing?"*

Watch For

Have students vote with thumbs up/down, then write their prediction and reasoning.

Slide 4: LEVER Framework

2 min

Say: *"We have five parts in our model today — two we control and three that respond. Let us figure out how they connect."*

Watch For

Preview all five components. Explain that Medium Density can go to ZERO, which represents the vacuum of space.

Slide 5: Activity 1: Components

8 min

Say: *"Which parts of this system do WE control? The person making the sound controls one thing, and the material controls another."*

Watch For

Guide sorting. Vibration Strength is external (the sound maker controls it). Medium Density is external (we choose what material sound travels through).

Slide 6: Activity 2: Connections

10 min

Say: *"This is tricky — some of these connections help sound and some work AGAINST sound. Let us figure out which is which."*

Watch For

Help students map all five relationships. Emphasize the negative connections and how they reduce loudness.

Slide 7: Activity 3: Simulation

12 min

Say: *"Now the big test — what happens when we set Medium Density to ZERO? Watch every single line on the graph."*

Watch For

Run air scenario first, then water, then set density to 0%. Students observe that ALL internal components drop to zero with no medium.

Slide 8: Discoveries

5 min

Say: *"The answer is clear — no molecules, no sound. Space movies have been lying to us this whole time!"*

Watch For

Create a class chart comparing sound in air vs. water vs. vacuum. Discuss why astronauts need radios.

Slide 9: STEM Challenge

12 min

Say: *"Now you are acoustic engineers. Which material carries sound the best from one side of the room to the other?"*

Watch For

Distribute materials (cups, string, wire, wooden dowels, plastic tubes). Groups build and test sound transmitters.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Vibration Strength, Medium Density
- **INTERNAL:** Wave Speed, Sound Loudness, Wave Absorption

Relationships

- **Vibration Strength to Sound Loudness:** POSITIVE (+) — A stronger vibration sends out sound waves with more energy (higher amplitude). More energy in the wave means louder sound when it reaches the listener, just like how shouting is louder than whispering.
- **Medium Density to Wave Speed:** POSITIVE (+) — In denser materials, molecules are packed closer together, so each molecule bumps into the next one faster. Sound travels about 4x faster in water than in air, and about 17x faster in steel than in air.
- **Medium Density to Wave Absorption:** NEGATIVE (-) — Denser materials actually absorb LESS sound energy as the wave passes through. In air, the loosely packed molecules scatter sound energy, absorbing it faster. In water and solids, the tightly packed molecules pass energy along more efficiently.
- **Wave Absorption to Sound Loudness:** NEGATIVE (-) — When more sound energy is absorbed by the medium, less energy reaches the listener, making the sound quieter. This is why whale songs can travel hundreds of miles in water (low absorption) but a shout barely carries a few hundred feet in air (higher absorption).
- **Vibration Strength to Wave Speed:** POSITIVE (+) — Stronger vibrations create waves with slightly more energy, which can push through a medium a bit faster. However, this effect is much smaller than the effect of medium density on wave speed.

Key Discoveries

- Sound MUST have a medium to travel through — in a vacuum (like space), there are no molecules to vibrate, so sound cannot exist
- Sound travels faster through denser materials — it moves about 4 times faster in water than in air, and about 15 times faster in steel than in air
- Denser media absorb less sound energy, so sound stays louder over longer distances in water and solids compared to air
- Both vibration strength and the medium affect loudness — a shout in water carries farther than a shout in air because water absorbs less energy

Multiple Choice Post-Assessment Answer Key

1. An astronaut in space watches a satellite explode 50 meters away. What does the astronaut hear?

Correct: C) Nothing at all, because sound cannot travel through the vacuum of space

Correct! Space is a vacuum with no molecules. Sound waves need molecules to vibrate and carry the wave from one place to another. With no medium, there is no sound at all.

2. In the model, Vibration Strength and Medium Density are external components. What makes them external?

Correct: B) They are conditions we can set or control, and the other components respond to them

Right! External components are inputs we can control. We decide how hard to hit a drum (Vibration Strength) and what material the sound travels through (Medium Density). The internal components respond to those choices.

3. The model shows that when Medium Density increases (from air to water to steel), Wave Speed also increases. Which best explains why?

Correct: B) In denser materials, molecules are packed closer together, so each molecule bumps into the next one faster

Exactly! In denser materials like water and steel, molecules are packed tightly together. When one molecule vibrates, it quickly bumps the next one, passing the sound wave along faster.

4. The relationship between Wave Absorption and Sound Loudness is NEGATIVE. What does this mean for how loud a sound is when it reaches you?

Correct: C) When absorption increases, loudness decreases because more sound energy is soaked up before reaching the listener

That is right! When a material absorbs more sound energy, less energy reaches your ears, so the sound is quieter. This is why whale songs travel far in water (low absorption) but shouts fade quickly in air (higher absorption).

5. A student sets Medium Density to 0% in the model, representing the vacuum of space. What happens to ALL the internal components?

Correct: C) All internal components drop to zero because there are no molecules to carry the sound wave

Correct! With no molecules at all, there is nothing to vibrate, nothing to carry a wave, and nothing to absorb. Every internal component drops to zero. This proves that sound absolutely requires a medium.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Sound can travel through empty space"	Sound absolutely CANNOT travel through a vacuum. Sound is a mechanical wave that requires molecules to vibrate. In space, there are virtually no molecules. This is different from light and radio waves, which ARE electromagnetic and can travel through vacuum. Every space movie showing loud explosions in space is scientifically wrong.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Sound travels at the same speed in all materials"	Sound speed changes dramatically depending on the medium. In air: 343 m/s. In water: 1,480 m/s. In steel: 5,960 m/s. Sound is FASTER in denser media because the molecules are closer together and transfer vibrations more quickly. Think of it like a line of people passing a ball — closer together means faster passing.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Louder sounds travel faster"	Volume (loudness) and speed are completely independent. A whisper and a scream both travel through air at 343 m/s. Loudness depends on amplitude (how big the vibrations are), while speed depends on the medium. A soft tap on steel travels at 5,960 m/s — much faster than a loud scream through air at 343 m/s.